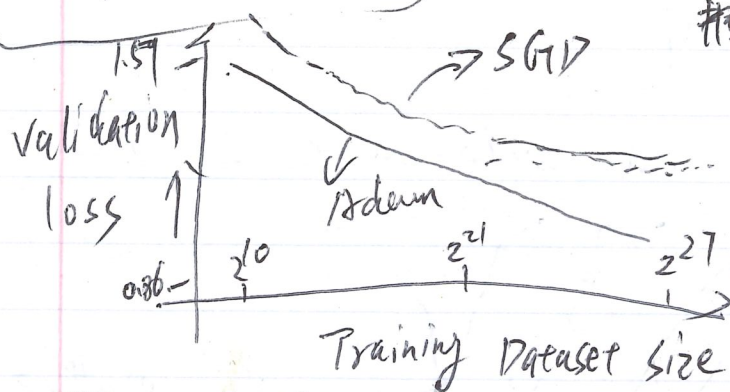


Cross-Architecture Scaling in Tay et al.

2) optimizer choice: what about (Adam vs SGD)



#3: Does depth or width make a huge difference?

↓ 1 vs 2 layers makes a huge difference
↓ more layers have ~~diminishing~~ diminishing returns below 10^7 params

3) Aspect ratio: width/length.

ratio of width to height of an image or video frame

↓ not all params are equal
params with embedding vs
params with non-embeddings

• No hyperparameters like aspect ratio depends on scale? how does scaling law interact with aspect ratio?

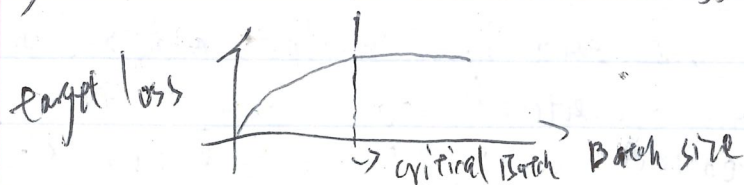
4) Batch size: critical batch size.

How does batch size interact with scaling

{ Batch size: known to have strong diminishing returns past a certain point.

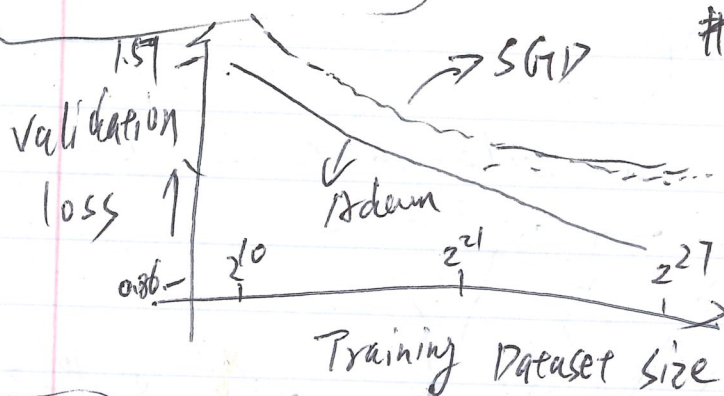
critical Batch: min number of examples for target loss
/ min number of steps for target loss

"The smaller the loss target, the bigger the batch"



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